Response to Editor and Reviewers

Manuscript: "JDCNet Uses Confidence Gated Privileged Modality Distillation for Cost Preserving X Ray Inference"
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We thank the Editor and the Reviewers for their careful assessment. We have revised the manuscript and respond below to every comment, point by point. Reviewer comments are reproduced in italics. Section, table, figure, and equation numbers refer to the revised manuscript (main text and Appendix A).

The Editor asked us to ensure that results are accurately reported, that overstated conclusions are rewritten, and that limitations are fully explained. Reviewer 6 raised closely related points about cross-validation, configuration selection, and statistical reporting. We responded in three steps.

1. We retrieved the complete per-run records of the original experiments from the original computing host and re-audited every number in the manuscript against them and against the training code. The audit led to substantive corrections, listed openly in Section 1 below.
2. We wrote an analysis plan, committed it to the code repository, and then ran a supplementary experiment on the original splits: 240 batch-matched supervised restarts, per-patient predictions for every checkpoint, and teacher-calibration diagnostics. This experiment directly addresses selection optimism, the training budget of the baseline, patient-level uncertainty, and the external test.
3. We rewrote the manuscript around the resulting evidence. No individual JDCNet configuration is significant after multiplicity adjustment. However, the training budget is unlikely to explain the results. When both methods select one of 16 runs per cell on the validation partition, JDCNet retains an advantage of +0.037 balanced accuracy (BA): 95% CI $[+0.016, +0.058]$ at the cell level and $[+0.007, +0.068]$ at the patient level. All models fail on an external set, and the conclusions are bounded accordingly.

Summary of principal changes

- **Selection history, direct tests, and a pre-registered nested comparison** (Sections 3.7, 4.4, and 4.7; Tables 2, 4, A1, and A6).
- **Training-budget confound tested.** The supervised baseline had been trained with batch size 512 and JDCNet with 128. Batch-matched restarts show no detectable effect of the batch size on supervised BA (0.620 versus 0.625; difference -0.005, CI $[-0.032, +0.022]$).
- **Complete absolute metrics** from the original records (Table 5), and **teacher reliability and gate coverage** recomputed from CT views re-rendered with the original code (Table 6). The re-rendered teachers reproduce the recorded teacher test BA exactly.
- **External test of the actual 510-patient models** with per-image predictions, image-level CIs, and paired DeLong tests (Section 4.10, Table 10).
- **New analysis of temperature scaling and hard targets** (Section 3.4, Eq. (4)), with the measured teacher calibration.

- **Corrected cohort description, implementation details, and comparator scope** (Sections 3.5, 4.1, and 4.2; Appendix A.3).
- **Moderated clinical and cost statements** (Sections 4.9 and 5; Table 11). "CT-only reference" replaces "teacher upper bound", and Macro-F1 is defined.
- **Literature updated** with 19 references from 2022–2026, including the work suggested by Reviewer 6 and all six works suggested by Reviewer 1.
- **Language and presentation.** Duplicated text is removed, spelling is unified to US English, acronyms are defined on first use, and the mechanism figure is redrawn (Fig. 1).

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1 Corrections to previously reported results

We regret that the previous version contained statements and numbers that are not supported by our records or by the training code. Each one has been corrected or removed.

1. **Pre-registration claim.** The previous version stated that the full JDCNet grid had been registered on a 226-patient pilot before the 510-patient cohort was unblinded. This was incorrect. The decision rule (mean ΔBA $\geq +0.03$ and lower CI bound > 0) was written in the analysis plan before any JDCNet run. The configurations, however, were explored in three stages on the 510-patient cohort, and that cohort had already been used for other comparisons. The two configurations that met the rule came from extension stages added after an initial eight-configuration grid had met it in none of its configurations. Section 3.7 now describes this chronology. Section 4.7 reports a nested comparison planned before it was run.
2. **Absolute performance table (Table 5).** The previous values were inconsistent with the recorded results. Every BA was about 0.021 too low; for example, the supervised baseline BA is 0.625, not 0.604. The AUC, sensitivity, and specificity values were also inconsistent. Table 5 is now computed from the per-run records.
3. **"Sensitivity gain" interpretation.** JDCNet's gains came through specificity, not sensitivity. The mean specificity change is positive for all 16 configurations, and the mean sensitivity change is negative for 12 (Table A1).
4. **Fold/seed win counts.** Eight of the 16 "Pos./15" entries in the sweep table were wrong. They are corrected from the records (Table 4, now reported as positive/tied/negative).
5. **BiomedCLIP control.** This control was evaluated on a 226-patient balanced subset, not on the 510-patient cohort. It is now reported with its correct cohort (Section 4.5).
6. **Comparators described as run "under the same splits".** Attention transfer, feature hints, and the module-augmented pilot were run only in an early small-cohort stage. Section 3.5 now states the cohort for every comparator.
7. **Comparator tuning budget.** On the 510-patient cohort, each logit-KD comparator was run with a single setting per teacher view (T=4, α=0.6), not the 4×4 grid previously stated. The previously reported same-modality distillation check has no record on any of our computing hosts and has been removed.
8. **Gate-coverage and teacher-calibration table.** The previous table included thresholds and teacher views for which no JDCNet runs exist, and its values could not be traced to any compu-

tation. The aggregate gate-behavior figure based on it has also been removed. We recomputed the diagnostics from CT views re-rendered with the original code (Table 6). The recomputed values differ substantially. For example, the gate retains 84–95% of training samples, not the 63–74% stated before, and the held-out teacher ECE is 0.17–0.19.

9. **Lightweight-backbone results (MobileNetV2, EfficientNet-B0).** No runs or code for these results exist on any of our computing hosts. They have been removed, and lightweight students are listed as future work. We apologize to Reviewer 3 of the previous round, to whom these results were presented as a response.

10. **Patient-level bootstrap table.** The original evaluation pipeline stored only aggregate metrics, so the previously reported patient-level intervals could not have been computed as described. We have now computed patient-level bootstrap intervals from newly generated per-patient predictions (Tables 2 and A6).

11. **Cross-source non-COVID control table.** This control used checkpoints from the early 19-patient study, not models trained "as in the main BIMCV protocol", and its values have no record. It has been removed.

12. **Deployment latency table.** The previous CPU, GPU, and edge (Jetson) timings and the 1.18× training overhead could not be traced to a measurement. We re-measured inference latency and memory with a released script and report them as illustrative (Table 9). The robust conclusion is structural: JDCNet and the supervised baseline share one deployed computation graph. The previous figure caption that quoted 0.094 M parameters for the ResNet-18 deployment path was wrong, and the figure has been redrawn.

13. **Cohort construction rules.** The previous appendix described a ±14-day pairing window, an earliest-pair rule, hashed-identifier splitting, and per-fold normalization statistics. None of these is implemented in the cohort-construction code. Appendix A.3 and Section 4.1 now describe the rules actually used, and two consequences are stated as limitations: pairs are same-patient but not necessarily same-visit, and the two classes come from two BIMCV releases.

14. **Implementation details.** The following are corrected in Section 4.2:
    - CT teachers were trained within every fold and seed, not "pre-trained once".
    - Checkpoints were selected on a validation fold.
    - Batch sizes differed between arms.
    - The initial 120-run JDCNet stage took about 32 minutes on two GPUs, not "6 hours".
    - The 3-slice view stacks three different slices.

15. **External test.** The previously reported external evaluation used students trained on a smaller BIMCV subset, not the 510-patient models. The revised Section 4.10 evaluates the actual 510-patient models and reaches the same conclusion.

16. **Gate-temperature rerun.** This rerun was performed on a different host, reports validation-partition BA, and partly used approximated CT views. It is now in Appendix A.8 with these caveats and is superseded by the supplementary experiment.

All cell-level statistics in the revised manuscript are computed by released scripts from the released per-run records. The patient-level, external, and teacher analyses additionally use per-image predictions, which are indexed by BIMCV and MIDRC identifiers and are therefore available from the

corresponding author on request rather than in the public repository.

Editor

Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained.

Response. We agree, and this comment shaped the revision.

- *Accurate reporting.* We audited every number against the original per-run records and the code; the corrections are listed in Section 1. Before the supplementary experiment, we committed its analysis plan to the repository and reported every pre-specified comparison (Section 4.7, Section 4.10, and Appendix Tables A6 and A7).
- *Overstated conclusions.* The paper no longer describes JDCNet as "validated" by a pre-specified gate. It now separates three findings:
  - individual configurations are not significant after multiplicity adjustment;
  - at the method level, JDCNet shows a small within-cohort advantage when its configuration is selected on validation data and compared with an equally tuned supervised baseline (+0.037 BA, patient-level CI $[+0.007, +0.068]$);
  - all models fail on an external set.

  We removed statements about triage, regulatory pathways, equal deployment cost beyond the computation graph, and the "two-thirds of the head-room" interpretation (Reviewer 6, Comments 4–5).
- *Limitations.* Section 4.12 covers:
  - selection and confirmatory status;
  - the training budget;
  - data and generalization, including the class-dependent mix of detector types in the training data;
  - task scope;
  - the mechanism, since the teachers supervise their own training patients;
  - reporting and statistics.

  Table 11 summarizes benefits and limitations.

Where in the manuscript: Abstract; Section 1.2; Sections 3.7 and 4.4–4.12; Sections 5–6; Tables 2, 4–6, 10, and 11; Appendix A.9.

Declaration Section: Authors are requested to include a separate section in the manuscript titled "Declarations". In this section, please clearly state the following items as individual subheadings: Ethical approval; Consent to participate; Consent to publish.

Response. The manuscript contains a section titled *Declarations* with the individual subheadings *Ethical approval*, *Consent to participate*, and *Consent to publish*. These are followed by Funding, Competing interests, Data availability, Materials availability, Code availability, and Author contribution. The ethical-approval statement identifies the approving committees of the original BIMCV

data collection (CEIm: 12/2020) and explains why no further approval was needed for this secondary analysis of de-identified public data. The consent statements explain that no patients were recruited or contacted and that the manuscript contains no identifiable personal data.

Where in the manuscript: Declarations (end of the main text).

Reviewer 1

1. Why is the evaluation only done on COVID-19 diagnosis and can this approach be extended to multi-class (multi-disease) problems?

Response. The study requires CT and chest radiographs of the *same* patients with labels. The only public cohort of this kind that was available to us is the BIMCV-COVID19 collection, which defines a binary COVID-19 versus non-COVID task. We now state this reason explicitly and treat the binary task as a limitation (Section 4.12, "Task scope").

Methodologically, the JDCNet objective extends directly to K mutually exclusive classes: the gate uses $c_i = \max_k p^{ct}_{i,k}$, and the hard or soft-KL auxiliary target is defined over all K classes. A new paragraph (Section 3.4, "Extension beyond two classes") explains two points. First, class-conditional thresholds may be needed for imbalanced label spaces, because the temperature–threshold equivalence of Eq. (4) no longer holds exactly for $K > 2$. Second, for multi-label chest radiograph findings the natural analog is a per-finding sigmoid teacher with a per-finding mask. We cite recent multi-label chest-radiograph work (CXR-LT, CheXzero) as the relevant setting. No multi-disease same-patient CT–X-ray cohort was available to us, so the paper states that this extension is untested.

Where in the manuscript: Section 3.4 ("Extension beyond two classes"); Section 4.12 ("Task scope"); Section 2.2.

2. How do you justify the performance drop of the model (BA approx 0.49) on the external data (MIDRC) and what is your solution to deal with domain changes?

Response. We do not treat the external result as something to justify away. In the revision we evaluate the actual 510-patient models with per-image predictions on the 126 external radiographs (Section 4.10, Table 10). Every model performs at chance, the supervised students as well as JDCNet: BA 0.49–0.51, AUC 0.49–0.51, and all 95% CIs include 0.5. Paired DeLong tests show no difference between JDCNet and supervised students, for example $\Delta\text{AUC} = -0.032$, CI $[-0.087, +0.023]$. Because AUC is also at chance, the failure is a loss of ranking signal, not a threshold offset.

We list possible causes without claiming to have identified one:

- *Acquisition differences.* We found a concrete candidate: in the training cohort, computed radiography makes up 62% of the positive and 45% of the negative radiographs ($p = 0.002$), whereas 85% of the external radiographs are digital radiography. Detector-related appearance is therefore associated with the training label but not with the external one.
- label-definition differences;
- a release-level source confound in the training data;
- prevalence and population differences;
- preprocessing differences.

The remedies we describe include multi-center training with site-held-out evaluation, single-domain generalization such as spectral adversarial augmentation (Zhang et al., AAAI 2023), acquisition-noise augmentation, harmonization or stratification of detector types, domain adaptation, and site-specific threshold recalibration. None of these remedies is evaluated here.

Where in the manuscript: Section 4.10 and Table 10; Section 4.12 ("Data and generalization").

3. Given the drastic changes in performance by moving the ζ_{gate} and λ parameters, how likely is overfitting on this dataset?

Response. We now answer this quantitatively (Section 4.4, "Sensitivity to τ_{gate} and λ, and risk of overfitting"; Section 4.7).

- *The changes across τ and λ are within the noise.* Moving τ_{gate} from 0.70 to 0.80 with the other settings fixed changes ΔBA by at most 0.021. The standard deviation of the 16 configuration means (0.011) is smaller than their typical standard error (0.014). For comparison, 16 supervised restarts that differed only in the random seed varied with a within-cell standard deviation of 0.041 BA. Run-to-run variability, not τ or λ, therefore dominates the spread of individual runs.
- *Model-level overfitting to the test folds is limited.* Checkpoints were selected on a separate validation fold, and CT teachers were trained within each fold.
- *Selection overfitting is real, and we quantified and controlled it.*
  - With directly computed paired t-tests, the two best configurations have $p = 0.014$ and 0.028; after Holm adjustment over 16 configurations these become 0.22 and 0.42. No single configuration is significant after adjustment.
  - In a pre-registered nested comparison, both JDCNet and supervised training selected one of 16 runs per cell on the validation partition, so no test data were used for selection. JDCNet retained an advantage of +0.037 BA (CI $[+0.016, +0.058]$; patient-level $[+0.007, +0.068]$).
  - Validation-based selection chose 11 different configurations across the 15 cells, so the advantage belongs to the configuration family rather than to one tuned setting.

Where in the manuscript: Sections 3.7, 4.4, and 4.7; Tables 2 and 4; Appendix Tables A1 and A6.

4. The manuscript relies heavily on sources before 2021. Consider this works: [six DOIs].

Response. Thank you for this suggestion. We updated the literature with 13 verified references from 2022–2026 that bear directly on the study:

- privileged-modality and missing-modality distillation in medical imaging (Chen et al., IEEE TMI 2022; Wang et al., MICCAI 2023);
- on-device chest-radiograph distillation (Termritthikun et al., IEEE/ACM TCBB 2024);
- confidence-aware and decoupled distillation (Zhang et al., ICASSP 2022; Zhao et al., CVPR 2022);
- distillation temperature (Li et al., AAAI 2023) and calibration of medical classifiers (Rajaraman et al., PLOS ONE 2022);
- multi-label chest-radiograph classification (Holste et al., Medical Image Analysis 2024; Tiu et al., Nature Biomedical Engineering 2022);
- external and prospective generalization of chest-radiograph models (Sun et al., Radiology: AI

2022; Ahluwalia et al., Radiology: AI 2023; Ong Ly et al., npj Digital Medicine 2024; Zhang et al., AAAI 2023).

We also cite the canonical temperature-scaling reference (Guo et al., ICML 2017) in response to Comment 6.

We have also added all six works suggested by the Reviewer, each cited where it connects to the argument:

- Gholinavaz et al., Scientific Reports 2025, on detector robustness under noise and weather augmentation, is cited among the candidate robustness remedies for the external shift (Section 4.12).
- Saeedi et al., IWINAC 2026, an audio–visual cross-attention fusion model, is cited as an example of multimodal fusion that requires all modalities at inference, which is the setting JDCNet avoids (Section 1.1).
- Nemati et al., Journal of Supercomputing 2025 (FER-HA), is cited in the discussion of attention-based models as a hybrid attention design with a moderate parameter count (Section 2.3).
- Shirini et al., Journal of Supercomputing 2025, on the effect of learning rate and training epochs, is cited where we discuss the training budget of the compared methods (Section 4.12).
- Shirini et al., Journal of Supercomputing 2024 (multi-objective NSGA-II), and Taheri Hajivand et al., Journal of Agricultural Machinery 2024 (metaheuristic time–cost balancing), are cited as tools for the multi-objective choice among configurations, with the caveat that such search must run inside a nested protocol to avoid selection optimism (Section 4.12).

Where in the manuscript: Sections 1.1, 2.1–2.8, 3.4, 4.10, and 4.12; reference list.

5. How much valuable uncertainty information is lost by replacing the teacher's soft distribution with hard argmax values?

Response. We answer this analytically and empirically (Section 3.4, "Information discarded by the hard target"; Section 4.4).

- *Analytically.* For two classes the teacher's distribution is fully described by its confidence c_i, and the non-target-class term of decoupled KD vanishes. For an admitted sample ($c_i > \tau$), replacing the distribution by its argmax label discards only the confidence magnitude above the threshold. The divergence between the one-hot target and the teacher distribution is below 0.36 nats at $\tau = 0.70$ and below 0.22 nats at $\tau = 0.80$; the discarded teacher entropy is below 0.88 and 0.72 bits. Below-threshold samples are excluded by either variant, so the gate itself is where the teacher's uncertainty is used.
- *Empirically.* At matched settings, the soft-KL target minus the hard target gives +0.028, +0.001, −0.003, and +0.010 BA (average +0.009), which is within the sampling uncertainty.
- *Mechanistically.* The new teacher diagnostics (Table 6) explain why so little is lost. On its own training patients, where the gate operates, the teacher is 94–96% accurate, and about 97% of admitted argmax targets equal the ground-truth label. The auxiliary target therefore acts mainly as CT-informed sample weighting, and the soft-KL variant additionally conveys the teacher's confidence.

Where in the manuscript: Section 3.4; Section 4.4 and Table 6; Table 4.

6. Why weren't corrective techniques such as Temperature Scaling used to address possible overconfidence in the teacher model before applying the confidence gate?

Response. We added an analysis and the corresponding measurements.

- *Temperature scaling of the gate is a change of threshold* (Section 3.4, Eq. (4)). For a two-class teacher with logit margin z_i, gating the temperature-scaled confidence $\sigma(|z_i|/T_g)$ at τ is identical to gating the raw confidence at $\tau' = \sigma(T_g \operatorname{logit}(\tau))$, and the argmax label is unchanged. Fitting a calibration temperature (Guo et al., 2017) therefore re-indexes τ rather than adding a new safeguard.
- *Measured calibration* (Table 6). On held-out patients the teachers are over-confident: validation and test ECE are 0.17–0.19, and the validation-fitted temperature is 1.7. With that temperature, the effective raw threshold of the mid-slice configuration would move from 0.80 to 0.92, and that of the 3-slice configuration from 0.70 to 0.81. The value 0.81 is close to the explored value 0.80, whereas 0.92 lies above both explored values (0.70 and 0.80). On the training partition, where the gate actually operates, the teachers are nearly calibrated (ECE 0.02–0.03), because they were fitted to those patients. Held-out calibration therefore does not describe the gate's operating regime.
- *What temperature scaling cannot do.* It cannot correct the ranking of teacher predictions: a confidently wrong prediction remains confidently wrong (Table 7, Case B). Cross-fitted teachers, class-conditional thresholds, and calibration beyond a single temperature are listed as open refinements (Section 4.12, "Mechanism").

Where in the manuscript: Section 3.4 (Eq. (4)); Section 4.4 and Table 6; Table 7; Section 4.12; Appendix A.8.

Reviewer 5

The authors have responded comprehensively to the editorial and reviewer feedback... Still to address: Reviewer 4, Comment 2: remove the near-verbatim duplication of the scaffold description between §§2.2 and 2.3. Reviewer 4, Comment 7: resolve remaining spelling inconsistencies (US/UK) and ensure all acronyms are defined on first use.

Response. We thank the Reviewer for the careful check of the previous rebuttal. Both outstanding items are addressed.

- *Duplication between Sections 2.2 and 2.3.* The scaffold and backbone description now appears only once, in Section 2.3. Section 2.2 was rewritten around chest-radiograph and CT classification and recent multi-label and prospective evaluations.
- *Spelling.* The manuscript now uses US English throughout. For example, "organise", "artefact", "prioritisation", "summarise", "favourable", "programme", and "analogue" were replaced or removed, and we checked the source files for remaining British forms.
- *Acronyms.* A new subsection (Section 3.1, "Notation and Abbreviations") defines every abbreviation used in the paper: AUC, BA, BIMCV, CE, CI, CNN, CPU, CT, DRR, ECE, GPU, KD, KL, MAC, MIDRC, SD, ViT, and wCE, together with the method names. Acronyms used before Section 3.1 are expanded at first use: CT, BA, CI, and KL in the abstract; KD and MAC in Section 2.1; ViT in Section 2.3. HU, FP16, and FP32 are expanded where they appear.

Because the Reviewer verified several items of our previous rebuttal against the manuscript, we must inform the Reviewer that our re-audit changed some of them (details in Section 1):

- *Reviewer 3, Comment 2 (qualitative gate visualization).* The aggregate gate figure has been removed. It is replaced by teacher-reliability diagnostics recomputed from re-rendered CT views (Table 6) and by the four case-level examples (Table 7).
- *Reviewer 3, Comment 3 (practical implications).* The interpretation is rewritten with corrected absolute values: the gain came through specificity, and no triage claim is made.
- *Reviewer 3, Comment 4 (lightweight models).* The MobileNetV2 and EfficientNet-B0 results have been removed because no runs exist on any of our hosts. Lightweight students are now listed as untested.
- *Reviewer 3, Comment 5, and Reviewer 4, Comment 4 (external validation).* The external test now evaluates the actual 510-patient models with per-image paired tests (Section 4.10).
- *Reviewer 4, Comment 3 (implementation details).* These are corrected: teachers were trained within each fold and seed, batch sizes differed between arms, and the wall time, hardware, and training overhead are restated (Section 4.2).
- *Reviewer 4, Comment 5 (statistics).* The CI-derived z/Cohen's-d columns are replaced by directly computed tests with multiplicity adjustment and by the pre-registered nested comparison (Tables 2, 4, A1, and A6).
- *Reviewer 4, Comment 6 (clinical significance).* This discussion is moderated (Section 5, Table 11).

We apologize for these errors in the previous version.

Where in the manuscript: Sections 2.2–2.3; Section 3.1; whole manuscript; Section 1 of this letter.

Reviewer 6

(1) Clarify whether CT teachers were trained separately within each cross-validation fold. The manuscript states that each CT teacher view was "pre-trained once" and that the frozen checkpoint was reused across student runs. It is unclear whether this means one teacher was trained within each training fold and reused only across seeds and student configurations belonging to that fold, or one teacher was trained on the complete 510-patient cohort and then used in all five cross-validation folds.

Response. Thank you for catching this ambiguity. The teachers were trained *within each fold*, so no held-out test patient was ever seen by a teacher. The procedure is as follows:

- For every fold and seed, a separate CT teacher for each view was trained only on that fold's three training folds, about 306 patients.
- Its checkpoint was selected on the fold's validation partition.
- It was then frozen and used only for the students of the same fold and seed.

This gives 15 teachers per view (five folds $\times$ three seeds). The phrase "pre-trained once" was inaccurate and has been removed. Section 4.2, Algorithm 1, Fig. 1, and Appendix A.3, rule (vi), now state the procedure explicitly; in each fold, fold k is the test partition and fold $(k{+}1) \bmod 5$ the validation partition.

One consequence of within-fold training is now analyzed explicitly (Section 4.4, Table 6). Because each teacher supervises the training patients on which it was itself fitted, its confidence on those patients is high, and the gate admits most of them.

Where in the manuscript: Section 4.2; Algorithm 1; Fig. 1; Appendix A.3; Section 4.4 and Table 6.

(2) The manuscript evaluates 16 JDCNet configurations on the same 510-patient cohort and designates the two configurations that satisfy the decision gate as validated operating points. The same cross-validation predictions are therefore used both to select the configurations and to estimate their performance. The Benjamini–Hochberg procedure may address one aspect of multiple hypothesis testing, but it does not remove the optimism introduced by selecting configurations and reporting their cross-validated performance on the same patients. Moreover, the manuscript does not provide the directly calculated p-values used for the correction.

Response. We agree with every part of this comment, and it led to the most important change in the revision. Our re-audit also showed that the configurations were explored in three stages on the same cohort (Section 1, item 1). We have withdrawn the designation "validated operating points" and added the following.

- *Chronology disclosed* (Section 3.7). Table 4 gives the exploration stage of every configuration.
- *Directly calculated p-values* (Table 4; Appendix Table A1). We retrieved the per fold/seed records of all 16 configurations. For each configuration we report the paired t-test, Wilcoxon signed-rank, bootstrap, and exact sign tests on the 15 cell differences.
  - The two selected configurations have $p_t = 0.014$ and 0.028.
  - Holm adjustment gives 0.22 and 0.42; Benjamini–Hochberg adjustment gives 0.22 for both.
  - No individual configuration is significant after adjustment.
- *Unselected averages.* The per-cell average over all 16 configurations is +0.020 (CI $[+0.003, +0.039]$, $p = 0.060$), and over the eight initial configurations +0.016 ($[-0.001, +0.035]$).
- *A pre-registered nested comparison with a matched control* (Section 4.7; Appendix A.9). Because Benjamini–Hochberg does not remove selection optimism, we ran a supplementary experiment designed for this question. Its analysis plan was committed to the code repository before any run. We trained 240 supervised restarts (16 per fold/seed cell) at the batch size used by JDCNet, on the original splits and with the original code.
  - *Design.* In every cell, JDCNet selected one of its 16 configurations and supervised training selected one of its 16 restarts, both by validation BA, and only the selected runs were evaluated on the test partition. Selection therefore never used test data, and the baseline received the same number of attempts.
  - *Primary result.* JDCNet exceeded supervised training by +0.037 BA (CI $[+0.016, +0.058]$, $p = 0.005$; 11 of 15 cells positive). A patient-level paired bootstrap gave $[+0.007, +0.068]$, $p = 0.017$.
  - *Matched control.* Best-of-16 selection improved the supervised baseline by only +0.013 (CI $[-0.011, +0.038]$).
  - *Training budget.* The batch-size difference identified in our audit is unlikely to explain the result: supervised BA was 0.620 at batch 128 versus 0.625 at batch 512.

- *Interpretation.* The revised conclusion is method-level and within-cohort. JDCNet shows a small advantage when its configuration is chosen on validation data and compared with an equally tuned baseline, whereas the performance of any particular configuration remains uncertain. We state that the experiment uses the same 510 patients, that the validation partition served both checkpoint and configuration selection (equally for both methods), and that an untouched cohort is still needed (Section 4.12).

Where in the manuscript: Abstract; Section 1.2; Sections 3.7, 4.4, and 4.7; Tables 2 and 4; Appendix Tables A1 and A6; Sections 4.12, 5, and 6.

(3) The external stress test is valuable and should remain prominent. However, the manuscript states that the three methods are "statistically indistinguishable" without reporting confidence intervals or a statistical comparison. Please report patient-level confidence intervals for BA and AUC and paired differences between each JDCNet model and the supervised baseline. This can be calculated from the existing predictions. The assertion that the failure reflects source shift "rather than a JDCNet-specific weakness" is plausible but not demonstrated. ... Acquisition protocol, label definition, prevalence, scanner, population, and preprocessing differences should be presented as possible explanations. A directly relevant reference ... Zhang J, Chao H, Dhurandhar A, Chen P-Y, Tajer A, Xu Y, Yan P. When Neural Networks Fail to Generalize? A Model Sensitivity Perspective. AAAI 2023.

Response. We agree, and the external test remains a dedicated subsection (Section 4.10) with a new table (Table 10).

- *The right models and per-image predictions.* The original evaluation pipeline stored only aggregate metrics, and the previously reported external evaluation had used students trained on a smaller BIMCV subset. We therefore re-ran inference with the 15 checkpoints of each 510-patient model and stored per-image probabilities for the 126 external radiographs (one per patient, 63/63).
- *Confidence intervals.* We report Wilson intervals for BA, which equals accuracy with balanced classes, and Hanley–McNeil intervals for AUC:
  - supervised (batch 512): BA 0.514 $[0.427, 0.599]$, AUC 0.505 $[0.404, 0.606]$;
  - 3-slice soft-KL JDCNet: BA 0.492 $[0.406, 0.578]$, AUC 0.498 $[0.397, 0.599]$;
  - mid-slice hard JDCNet: BA 0.503 $[0.417, 0.589]$, AUC 0.511 $[0.409, 0.612]$.

  Every interval includes 0.5.
- *Paired differences.* Paired DeLong tests on the checkpoint-averaged per-image scores give $\Delta\text{AUC} = -0.032$ (CI $[-0.087, +0.023]$, $p = 0.26$) for 3-slice soft-KL and -0.009 ($[-0.065, +0.047]$, $p = 0.75$) for mid-slice hard against the batch-512 supervised students. The mean cell-wise BA differences are within ± 0.03, with CIs including zero. The pre-specified McNemar tests per checkpoint pair gave $p < 0.05$ in 2 of 15 pairs against the batch-512 students, both favoring the supervised model, and in none of 15 against the batch-128 students. The phrase "statistically indistinguishable" has been removed; the text states that the data show no detectable difference.
- *Causes presented as possibilities.* We removed "rather than a JDCNet-specific weakness". The text now states that the data show that all tested models fail, not why, and lists acquisition, label-definition, source-confound, prevalence/population, and preprocessing differences. For acquisition we report a concrete candidate: detector type is associated with the label in the training data (computed radiography in 62% of positives versus 45% of negatives, $p = 0.002$), while 85%

of the external radiographs are digital radiography.

- *Reference added.* We cite Zhang et al. (AAAI 2023) for the gap between internal validation and robustness after single-source training, and mention spectral adversarial augmentation as a candidate remedy (Sections 2.8, 4.10, and 4.12). We also cite a prospective multi-hospital evaluation of a COVID-19 chest-radiograph model (Sun et al., 2022) and a shortcut-learning analysis (Ong Ly et al., 2024).

Where in the manuscript: Section 4.10 and Table 10; Sections 2.8 and 4.12.

(4) The manuscript is generally cautious, but the following statements should be further moderated. "X-ray triage or prioritisation" is not evaluated as a clinical workflow. Matching parameter count, MACs, and single-image latency does not establish equal end-to-end deployment cost, calibration, safety, or clinical utility. A higher retrospective sensitivity does not by itself demonstrate better triage. Describing the software as following the "same premarket evaluation pathway" involves regulatory assumptions outside the presented experiment. Please restrict the conclusion to the demonstrated result: the deployed computation graph is identical to the supervised ResNet-18 baseline and therefore has no incremental model-inference cost under the reported benchmark.

Response. We have made all four moderations and restricted the conclusion as requested.

- *Triage or prioritization.* Every mention of triage or prioritization as a benefit has been removed. Section 5 states that no triage or prioritization workflow was evaluated.
- *Deployment cost.* Section 4.9 states that, because the JDCNet and supervised students share one computation graph, they have identical parameters, MACs, and per-image latency, and that this "does not establish equal end-to-end deployment cost, calibration, safety, or clinical utility".
- *Sensitivity and triage.* JDCNet's gains came through specificity, while sensitivity was unchanged or lower (Section 4.4, Table 5, Table A1). The text states that "a higher BA at this retrospective operating point does not by itself indicate better clinical triage", and Section 5 repeats that a higher retrospective BA does not demonstrate better triage.
- *Regulatory pathway.* The "same premarket evaluation pathway" statement has been removed. Section 5 says only that clinical use would require prospective evaluation against patient outcomes and assessment under the applicable regulatory framework.
- *Conclusion.* The conclusion opens with the demonstrated result: the deployed X-ray classifier has the same computation graph as the supervised baseline and therefore adds no model-inference cost.

Where in the manuscript: Sections 4.4, 4.9, 5, and 6; Table 11.

(5) The "CT teacher upper bound" is not a mathematical upper bound on student performance. "Privileged-modality reference" or "CT-only reference" would be more accurate. The statement that the two passing configurations recover "roughly two-thirds of the available gap" is a ratio of uncertain and selected estimates. Either provide uncertainty for this ratio or describe it explicitly as a descriptive calculation. Clarify whether the reported F1 is binary F1 or macro-F1. Table 5 labels it only as "F1."

Response.

- *Terminology.* "Teacher upper bound" is replaced by "CT-only reference" throughout the paper, the tables, and the appendix. Sections 3.1 and 4.6 state explicitly that it is not a mathematical upper bound on the student.

- *The "two-thirds" ratio.* The phrase has been removed from the abstract, contributions, and discussion. Section 4.4 gives the ratio only as an explicitly descriptive calculation without uncertainty: the selected estimates correspond to 0.68 and 0.73 of the CT-only reference differences, and the 16-configuration average to 0.41. The text adds that these ratios are not estimates of the transferable fraction of CT signal.
- *F1.* The reported F1 is macro-F1, the unweighted mean of the per-class F1 scores. Table 5 labels the column "Macro-F1", and Section 3.1 defines it.

Where in the manuscript: Sections 3.1, 4.4, and 4.6; Tables 2, 5, and A2.

Summary of files changed

The revised clean manuscript (main text and appendix), the bibliography, the separate title page, the standalone tables file, and this point-by-point response have been updated and compiled. A marked-up version of the main text against the previous submission is provided. The analysis plan of the supplementary experiment, the scripts, and the per-run aggregate records used to compute the cell-level statistics are released with the code; the per-image predictions used for the patient-level, external, and teacher analyses are available on request, as stated in the Data availability declaration.